

2 Announcements

We will be releasing HW1 today

- It is due in 2 weeks (1/23 at 11:59 PM)
- The homework is long
 - Requires proving theorems as well as coding
- **Please start early**

Recitation sessions:

- **Spark Tutorial:**
Friday, 3:00-4:20pm in Skilling Auditorium

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Frequent Itemset Mining & Association Rules

CS246: Mining Massive Datasets
Jure Leskovec, Stanford University
<http://cs246.stanford.edu>



Association Rule Discovery

Supermarket shelf management – Market-basket model:

- **Goal:** Identify items that are bought together by sufficiently many customers
- **Approach:** Process the sales data collected with barcode scanners to find dependencies among items
- **A classic rule:**
 - If someone buys diaper and milk, then he/she is likely to buy beer
 - Don't be surprised if you find six-packs next to diapers!

The Market-Basket Model

- A large set of **items**
 - e.g., things sold in a supermarket
- A large set of **baskets**
 - Each basket is a **small subset of items**
 - e.g., the things one customer buys on one day
- **Discover association rules:**

People who bought $\{x,y,z\}$ tend to buy $\{v,w\}$

 - Example application: Amazon

Input:

<i>Basket</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Output:

Rules Discovered:

$\{\text{Milk}\} \rightarrow \{\text{Coke}\}$

$\{\text{Diaper, Milk}\} \rightarrow \{\text{Beer}\}$

More generally

- **A general many-to-many mapping (association) between two kinds of things**
 - But we ask about connections among “items”, not “baskets”
- **Items and baskets are abstract:**
 - **For example:**
 - Items/baskets can be products/shopping basket
 - Items/baskets can be words/documents
 - Items/baskets can be basepairs/genes
 - Items/baskets can be drugs/patients

Applications – (1)

- **Items** = products; **Baskets** = sets of products someone bought in one trip to the store
- **Real market baskets:** Chain stores keep TBs of data about what customers buy together
 - Tells how typical customers navigate stores, lets them position tempting items together:
 - Apocryphal story of “diapers and beer” discovery
 - Used to position potato chips between diapers and beer to enhance sales of potato chips
- **Amazon’s ‘people who bought X also bought Y’**

Applications – (2)

- **Baskets** = sentences; **Items** = documents in which those sentences appear
 - Items that appear together too often could represent plagiarism
 - Notice items do not have to be “in” baskets
- **Baskets** = patients; **Items** = drugs & side-effects
 - Has been used to detect combinations of drugs that result in particular side-effects
 - **But requires extension:** Absence of an item needs to be observed as well as presence

Outline

First: Define

Frequent itemsets

Association rules:

Confidence, Support, Interestingness

Then: Algorithms for finding frequent itemsets

Finding frequent pairs

A-Priori algorithm

PCY algorithm

Frequent Itemsets

- **Simplest question:** Find sets of items that appear together “frequently” in baskets
- **Support** for itemset I : Number of baskets containing all items in I
 - (Often expressed as a fraction of the total number of baskets)
- Given a **support threshold s** , then sets of items that appear in at least s baskets are called **frequent itemsets**

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Support of
 $\{\text{Beer, Bread}\} = 2$

Example: Frequent Itemsets

- **Items** = {milk, coke, pepsi, beer, juice}
- **Support threshold** = 3 baskets

$$B_1 = \{m, c, b\}$$

$$B_2 = \{m, p, j\}$$

$$B_3 = \{m, b\}$$

$$B_4 = \{c, j\}$$

$$B_5 = \{m, p, b\}$$

$$B_6 = \{m, c, b, j\}$$

$$B_7 = \{c, b, j\}$$

$$B_8 = \{b, c\}$$

- **Frequent itemsets:** {m}, {c}, {b}, {j},
{m,b} , {b,c} , {c,j}.

Define: Association Rules

- **Define: Association Rules:**

If-then rules about the contents of baskets

- $\{i_1, i_2, \dots, i_k\} \rightarrow j$ means: “if a basket contains all of $i_1, \dots, i_k$ then it is *likely* to contain j ”
- In practice there are many rules, want to find significant/interesting ones!
- Confidence of association rule is the probability of j given $I = \{i_1, \dots, i_k\}$

$$\text{conf}(I \rightarrow j) = \frac{\text{support}(I \cup j)}{\text{support}(I)}$$

Interesting Association Rules

- **Not all high-confidence rules are interesting**
 - The rule $X \rightarrow \textit{milk}$ may have high confidence for many itemsets X , because milk is just purchased very often (independent of X) and the confidence will be high
- **Interest of an association rule $I \rightarrow j$:**
abs. difference between its confidence and the fraction of baskets that contain j
$$\textit{Interest}(I \rightarrow j) = |\textit{conf}(I \rightarrow j) - \text{Pr}[j]|$$
 - Interesting rules are those with high interest values (usually above 0.5)

Example: Confidence and Interest

$$B_1 = \{m, c, b\}$$

$$B_2 = \{m, p, j\}$$

$$B_3 = \{m, b\}$$

$$B_4 = \{c, j\}$$

$$B_5 = \{m, p, b\}$$

$$B_6 = \{m, c, b, j\}$$

$$B_7 = \{c, b, j\}$$

$$B_8 = \{b, c\}$$

■ Association rule: $\{m, b\} \rightarrow c$

- Support = 2

- Confidence = $2/4 = 0.5$

- Interest = $|0.5 - 5/8| = 1/8$

- Item c appears in $5/8$ of the baskets

- The rule is not very interesting!

Association Rule Mining

- **Problem:** Find all association rules with support $\geq s$ and confidence $\geq c$
 - **Note:** Support of an association rule is the support of the set of items in the rule (left and right side)
- **Hard part:** Finding the frequent itemsets!
 - If $\{i_1, i_2, \dots, i_k\} \rightarrow j$ has high support and confidence, then both $\{i_1, i_2, \dots, i_k\}$ and $\{i_1, i_2, \dots, i_k, j\}$ will be “frequent”

$$\text{conf}(I \rightarrow j) = \frac{\text{support}(I \cup j)}{\text{support}(I)}$$

Mining Association Rules

- **Step 1:** Find all frequent itemsets I
 - (we will explain this next)
- **Step 2: Rule generation**
 - For every subset A of I , generate a rule $A \rightarrow I \setminus A$
 - Since I is frequent, A is also frequent
 - **Variant 1:** Single pass to compute the rule confidence
 - $\text{confidence}(A, B \rightarrow C, D) = \text{support}(A, B, C, D) / \text{support}(A, B)$
 - **Variant 2:**
 - **Observation:** If $A, B, C \rightarrow D$ is below confidence, then so is $A, B \rightarrow C, D$
 - Can generate “bigger” rules from smaller ones!
 - **Output the rules above the confidence threshold**

$$\text{conf}(I \rightarrow j) = \frac{\text{support}(I \cup j)}{\text{support}(I)}$$

Example

$$B_1 = \{m, c, b\}$$

$$B_2 = \{m, p, j\}$$

$$B_3 = \{m, c, b, n\}$$

$$B_4 = \{c, j\}$$

$$B_5 = \{m, p, b\}$$

$$B_6 = \{m, c, b, j\}$$

$$B_7 = \{c, b, j\}$$

$$B_8 = \{b, c\}$$

- Support threshold $s = 3$, confidence $c = 0.75$

- Step 1) Find frequent itemsets:

- $\{b, m\} \{b, c\} \{c, m\} \{c, j\} \{m, c, b\}$

- Step 2) Generate rules:

- ~~$b \rightarrow m: c=4/6$~~ $b \rightarrow c: c=5/6$ ~~$b, c \rightarrow m: c=3/5$~~
- $m \rightarrow b: c=4/5$... $b, m \rightarrow c: c=3/4$
- ~~$b \rightarrow c, m: c=3/6$~~

Compacting the Output

- To reduce the number of rules, we can post-process them and only output:

- **Maximal frequent itemsets:**

No immediate superset is frequent

- Gives more pruning

or

- **Closed itemsets:**

No immediate superset has the same support (> 0)

- Stores not only frequent information, but exact supports/counts

Example: Maximal/Closed

	Support	Maximal(s=3)	Closed	
A	4	No	No	
B	5	No	Yes	Frequent, but superset BC also frequent.
C	3	No	No	
AB	4	Yes	Yes	Frequent, and its only superset, ABC, not freq.
AC	2	No	No	
BC	3	Yes	Yes	Superset BC has same support.
ABC	2	No	Yes	Its only super- set, ABC, has smaller support.

Step 2: Finding Frequent Itemsets

Itemsets: Computation Model

- **Back to finding frequent itemsets**
- Typically, data is kept in flat files rather than in a database system:
 - Stored on disk
 - Stored basket-by-basket
 - Baskets are **small** but we have many baskets and many items
 - Expand baskets into pairs, triples, etc. as you read baskets
 - Use **k** nested loops to generate all sets of size **k**

Item
Item
Item
Item
Item
Item
Item
Item
Item
Item
Item
Etc.

Items are positive integers, and boundaries between baskets are -1.

Note: We want to find frequent itemsets. To find them, we have to count them. To count them, we have to enumerate them.

Computation Model

- The true cost of mining disk-resident data is usually the **number of disk I/Os**
- In practice, association-rule algorithms read the data in *passes*
 - all baskets read in turn
- We measure the cost by the **number of passes** an algorithm makes over the data

Item
Item
Item
Item
Item
Item
Item
Item
Item
Item
Item
Item
Etc.

Items are positive integers, and boundaries between baskets are -1.

Main-Memory Bottleneck

- For many frequent-itemset algorithms, **main-memory** is the critical resource
 - As we read baskets, we need to count something, e.g., occurrences of pairs of items
 - The number of different things we can count is limited by main memory
 - Swapping counts in/out is a disaster

Finding Frequent Pairs

- The hardest problem often turns out to be finding the frequent **pairs** of items $\{i_1, i_2\}$
 - **Why?** Freq. pairs are common, freq. triples are rare
 - **Why?** Probability of being frequent drops exponentially with size; number of sets grows more slowly with size
- **Let's first concentrate on pairs, then extend to larger sets**

Finding Frequent Pairs

■ The approach:

- We always need to “generate” all the itemsets
- But we would only like to count (keep track of) those itemsets that in the end turn out to be frequent

■ Scenario:

- Imagine we aim to identify frequent pairs
- We will need to enumerate all pairs of items
 - For every basket, enumerate all pairs of items in that basket
- **But**, rather than keeping a count for every pair, we hope to discard a lot of pairs and only keep track of the ones that will in the end turn out to be frequent

Naïve Algorithm

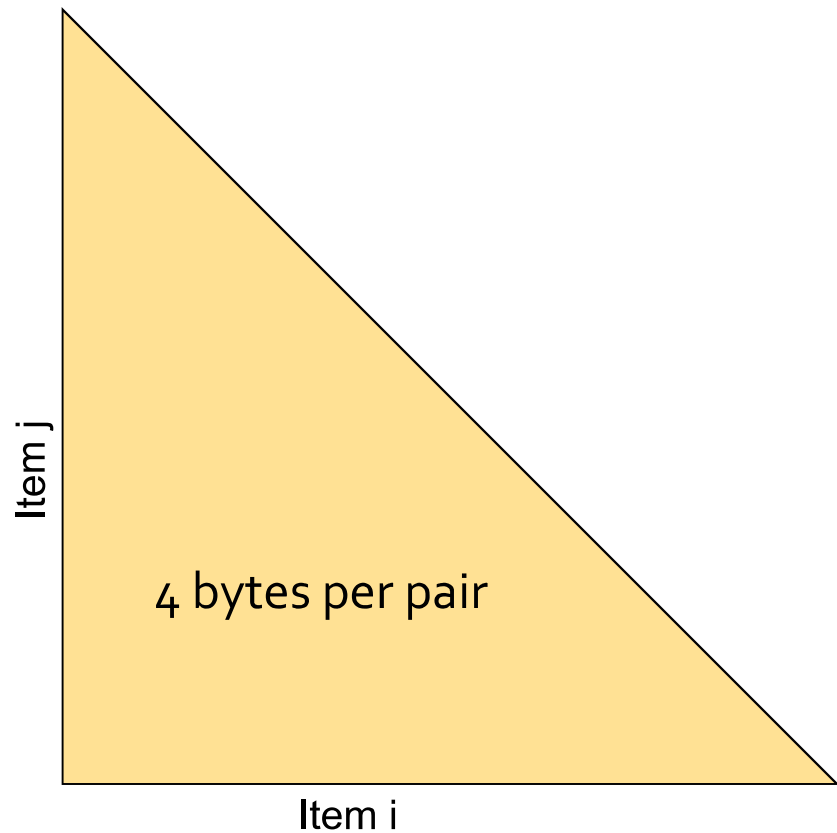
- **Naïve approach to finding frequent pairs**
- Read file once, counting in main memory the occurrences of each pair:
 - From each basket b of n_b items, generate its $n_b(n_b-1)/2$ pairs by two nested loops
 - A data structure then keeps count of every pair
- **Fails if $(\#items)^2$ exceeds main memory**
 - **Remember:** $\#items$ can be 100K (Wal-Mart) or 10B (Web pages)
 - Suppose 10^5 items, counts are 4-byte integers
 - Number of pairs of items: $10^5(10^5-1)/2 \approx 5 \cdot 10^9$
 - Therefore, $2 \cdot 10^{10}$ (20 gigabytes) of memory is needed

Counting Pairs in Memory

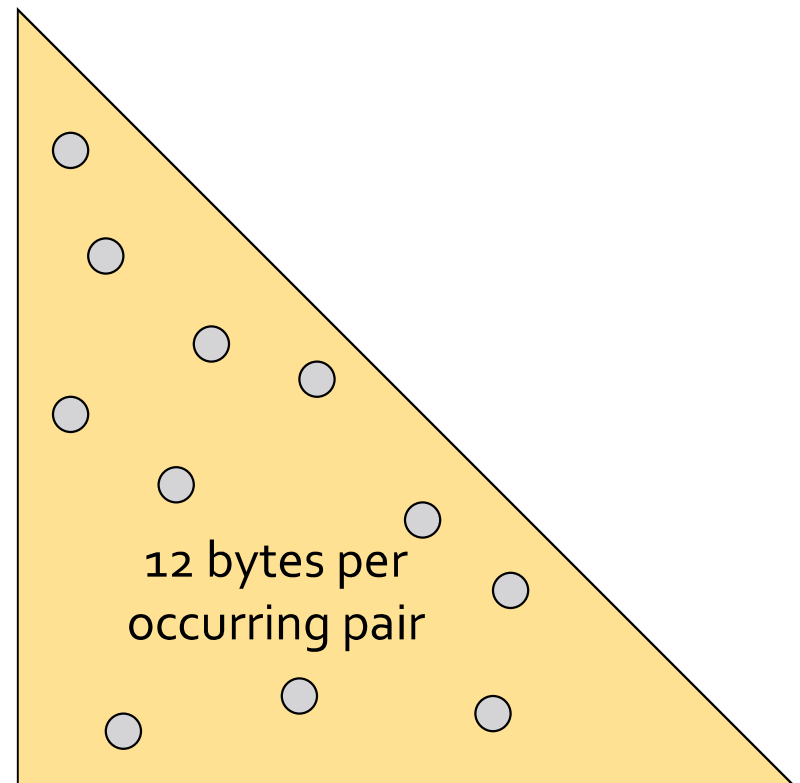
Goal: Count the number of occurrences of each pair of items (i,j):

- **Approach 1:** Count all pairs using a matrix
- **Approach 2:** Keep a table of triples $[i, j, c]$ = “the count of the pair of items $\{i, j\}$ is c .”
 - If integers and item ids are 4 bytes, we need approximately 12 bytes for pairs with count > 0
 - Plus some additional overhead for the hashtable

Comparing the Two Approaches



Triangular Matrix

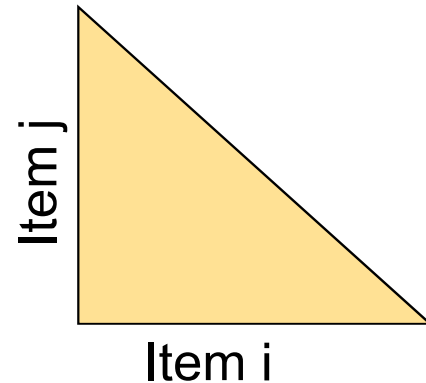


Triples (item i, item j, count)

Comparing the Two Approaches

■ Approach 1: Triangular Matrix

- n = total number items
- Count pair of items $\{i, j\}$ only if $i < j$
- Keep pair counts in lexicographic order:
 - $\{1,2\}, \{1,3\}, \dots, \{1,n\}, \{2,3\}, \{2,4\}, \dots, \{2,n\}, \{3,4\}, \dots$
- Pair $\{i, j\}$ is at position: $[n(n-1) - (n-i)(n-i+1)]/2 + (j-i)$
- Total number of pairs $n(n-1)/2$; total bytes = $O(n^2)$
- **Triangular Matrix** requires 4 bytes per pair



- **Approach 2** uses **12 bytes** per occurring pair (*but only for pairs with count > 0*)
- Approach 2 beats Approach 1 if less than **1/3** of possible pairs actually occur

Comparing the Two Approaches

■ Approach 1: Triangular Matrix

- n = total number items

- Correlation

- Key

-

- Pair

- Total

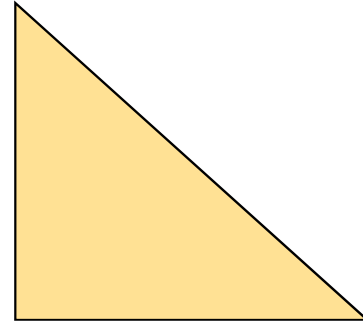
- Triangular

Problem is if we have too many items so all the pairs do not fit into memory.

Can we do better?

- Approach 2 (but)

- Approach 2 beats Approach 1 if less than **1/3** of possible pairs actually occur



$$[j(j-1)]/2 + (j-i)$$
$$O(n^2)$$

pair

A-Priori Algorithm

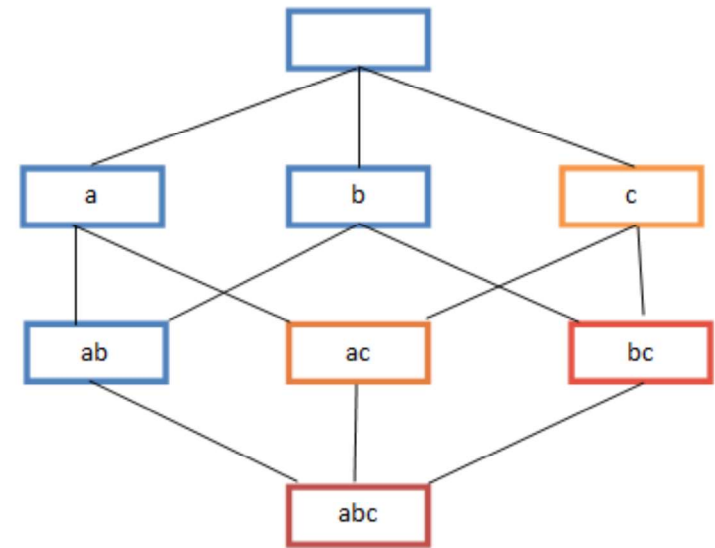
Key concepts:

- Monotonicity of “Frequent”
- Notion of Candidate Pairs
- Extension to Larger Itemsets

A-Priori Algorithm – (1)

- A **two-pass** approach called *A-Priori* limits the need for main memory
- **Key idea:** *monotonicity*
 - If a set of items I appears at least s times, so does every **subset** J of I
- **Contrapositive for pairs:**

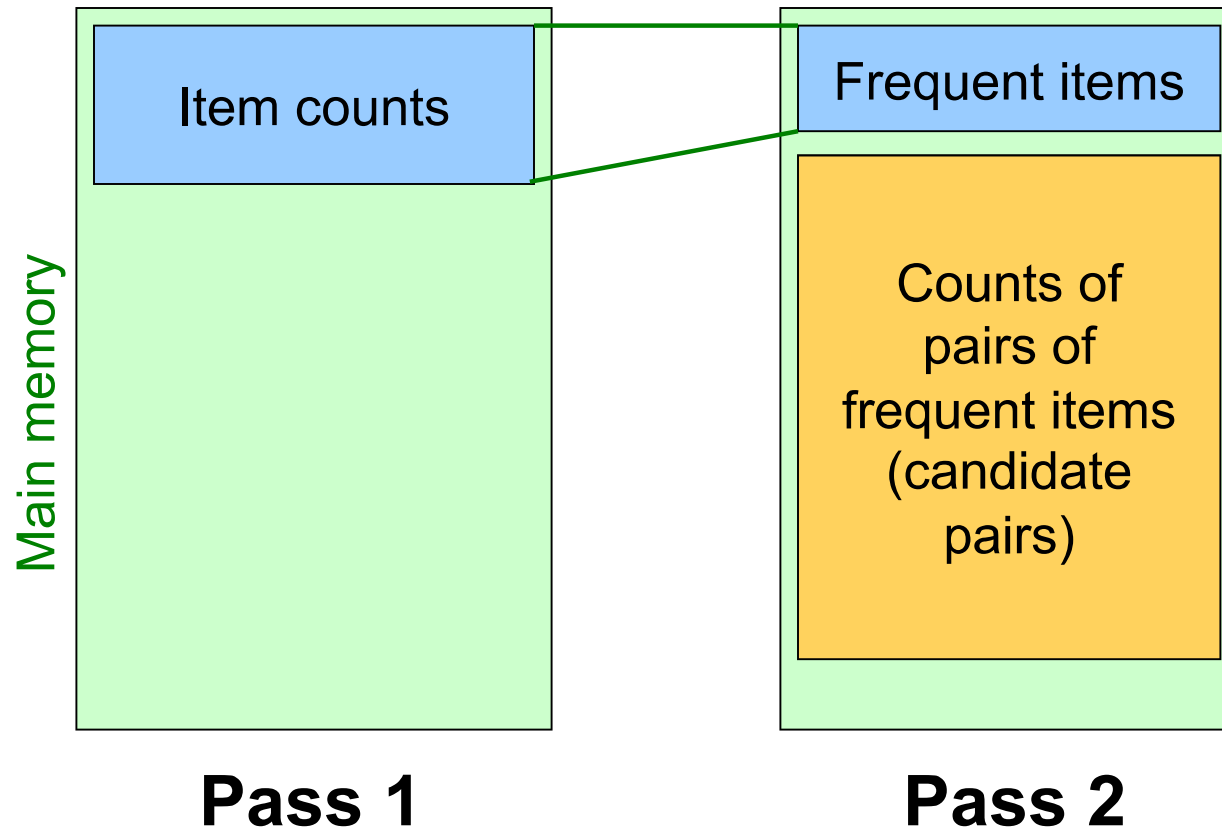
If item i does not appear in s baskets, then no pair including i can appear in s baskets
- **So, how does A-Priori find freq. pairs?**



A-Priori Algorithm – (2)

- **Pass 1:** Read baskets and count in main memory the # of occurrences of each **individual item**
 - Requires only memory proportional to #items
- **Items that appear $\geq s$ times are the frequent items**
- **Pass 2:** Read baskets again and keep track of the count of only those pairs where both elements are frequent (from Pass 1)
 - Requires memory (for counts) proportional to square of the number of **frequent** items (not the square of total # of items)
 - Plus a list of the frequent items (so you know what must be counted)

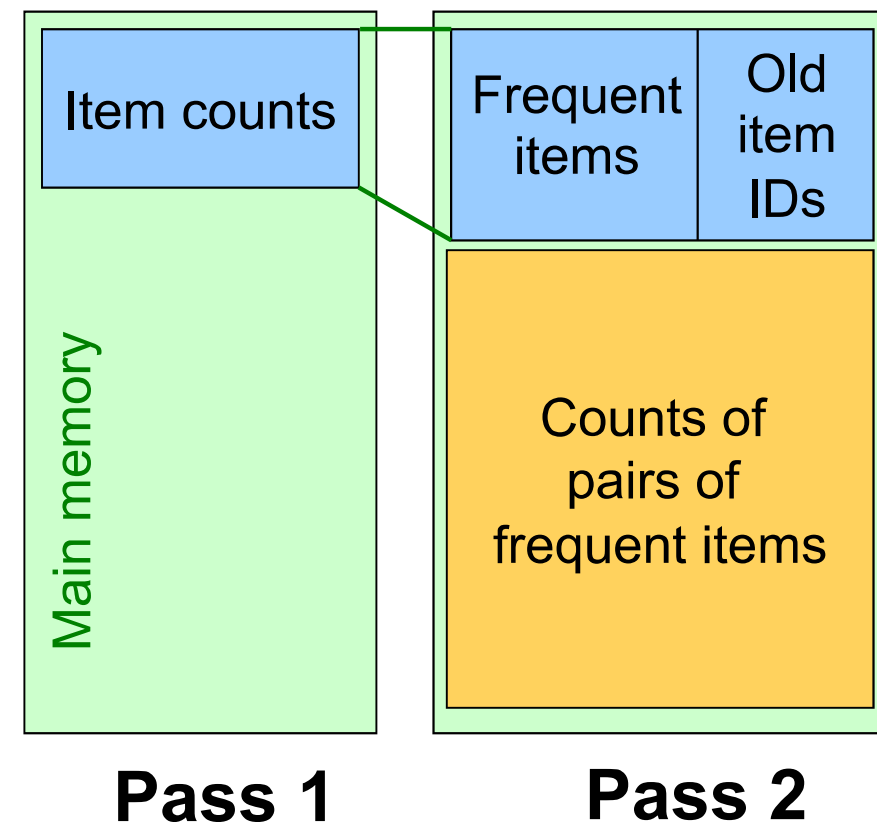
Main-Memory: Picture of A-Priori



Green box represents the amount of available main memory. Smaller boxes represent how the memory is used.

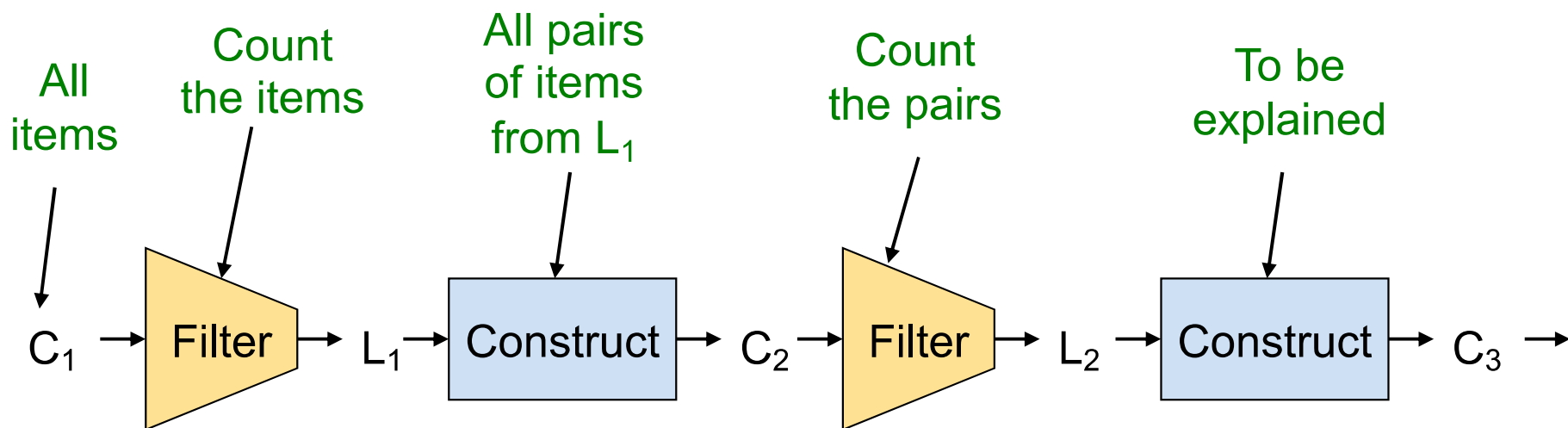
Detail for A-Priori

- You can use the triangular matrix method with n = number of frequent items
 - May save space compared with storing triples
- **Trick:** re-number frequent items 1,2,... and keep a table relating new numbers to original item numbers



Frequent Triples, Etc.

- For each k , we construct two sets of *k -tuples* (sets of size k):
 - C_k = *candidate k -tuples* = those that might be frequent sets (support $\geq s$) based on information from the pass for $k-1$
 - L_k = the set of truly frequent k -tuples



Example

** Note here we generate new candidates by generating C_k from L_{k-1} and L_1 .

But one can be more careful with candidate generation. For example, in C_3 we know $\{b,m,j\}$ cannot be frequent since $\{m,j\}$ is not frequent

■ Hypothetical steps of the A-Priori algorithm

- $C_1 = \{ \{b\} \{c\} \{j\} \{m\} \{n\} \{p\} \}$
- Count the support of itemsets in C_1
- Prune non-frequent. We get: $L_1 = \{ b, c, j, m \}$
- Generate $C_2 = \{ \{b,c\} \{b,j\} \{b,m\} \{c,j\} \{c,m\} \{j,m\} \}$
- Count the support of itemsets in C_2
- Prune non-frequent. $L_2 = \{ \{b,m\} \{b,c\} \{c,m\} \{c,j\} \}$
- Generate $C_3 = \{ \{b,c,m\} \{b,c,j\} \{b,m,j\} \{c,m,j\} \}$
- Count the support of itemsets in C_3
- Prune non-frequent. $L_3 = \{ \{b,c,m\} \}$

**

A-Priori for All Frequent Itemsets

- One pass for each k (itemset size)
- Needs room in main memory to count each candidate k -tuple
- For typical market-basket data and reasonable support (e.g., 1%), $k = 2$ requires the most memory
- **Many possible extensions:**
 - Association rules with intervals:
 - For example: Men over 65 have 2 cars
 - Association rules when items are in a taxonomy
 - Bread, Butter $\rightarrow$ FruitJam
 - BakedGoods, MilkProduct $\rightarrow$ PreservedGoods
 - Lower the support s as itemset gets bigger

PCY (Park-Chen-Yu) Algorithm

Key concepts:

- Improvement to A-Priori
- Exploits Empty Memory on First Pass
- Frequent Buckets

PCY (Park-Chen-Yu) Algorithm

■ **Observation:**

In pass 1 of A-Priori, most memory is idle

- We store only individual item counts
- **Can we use the idle memory to reduce memory required in pass 2?**

■ **Pass 1 of PCY:** In addition to item counts, maintain a hash table with as many **buckets** as fit in memory

Note:
Bucket $\neq$ Basket

- Keep a **count** for each bucket into which **pairs** of items are **hashed**
 - **For each bucket just keep the count, not the actual pairs that hash to the bucket!**

PCY Algorithm – First Pass

```
FOR (each basket) :  
    FOR (each item in the basket) :  
        add 1 to item's count;  
New in PCY { FOR (each pair of items in the basket) :  
                hash the pair to a bucket;  
                add 1 to the count for that bucket;
```

■ Few things to note:

- Pairs of items need to be generated from the input file; they are not present in the file
- We are not just interested in the presence of a pair, but we need to see whether it is present at least s (support) times

Observations about Buckets

- **Observation:** If a bucket contains a **frequent pair**, then the bucket is surely **frequent**
- However, even without any frequent pair, a bucket can still be frequent 😞
 - So, we cannot use the hash to eliminate any member (pair) of a “frequent” bucket
- **But, for a bucket with total count less than s , none of its pairs can be frequent 😊**
 - Pairs that hash to this bucket can be eliminated as candidates (even if the pair consists of 2 frequent items)
- **Pass 2:**
Only count pairs that hash to frequent buckets

PCY Algorithm – Between Passes

- **Replace the buckets by a bit-vector:**
 - **1** means the bucket count exceeded the support s (call it a **frequent bucket**); **0** means it did not
- **4-byte integer counts are replaced by bits, so the bit-vector requires 1/32 of memory**
- **Also, decide which items are frequent and list them for the second pass**

PCY Algorithm – Pass 2

- Count all pairs $\{i, j\}$ that meet the conditions for being a **candidate pair**:
 1. Both i and j are frequent items
 2. The pair $\{i, j\}$ hashes to a bucket whose bit in the bit vector is **1** (i.e., a **frequent bucket**)
- **Both conditions are necessary for the pair to have a chance of being frequent**

Main-Memory: Picture of PCY

